

FLY ASH BRICK PROJECT: FEASIBILITY STUDY USING CVP ANALYSIS

S. K. Mitra and Shubhra Hajela wrote this case solely to provide material for class discussion. The authors do not intend to illustrate either effective or ineffective handling of a managerial situation. The authors may have disguised certain names and other identifying information to protect confidentiality.

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In April 2013, Rajiv Sharma had a business plan. For quite some time, he had been doing some market research exploring the potential of his “dream project” — establishing a fly ash brick manufacturing unit — and had seen huge potential for profit in the project. His long-time friend Alok Gupta was also interested but had a few doubts regarding the feasibility of the project. Sharma, on the other hand, was sure of his plan. Years of work in the construction industry had shown him the potential of using large volumes of fly ash bricks in construction, especially in housing and infrastructure projects.

On the basis of preliminary analysis, he decided to set up a plant that would have the capacity to manufacture four million bricks per year. Though actual production would depend on market demand, the partners estimated that 2.4 million bricks could be sold per year at an average selling price of Rs7,000¹ per 1,000 bricks. He wanted to ascertain the feasibility of the project using a cost-volume-profit (CVP) analysis.

BACKGROUND

Fly ash was a residue obtained after combustion of coal. India utilized huge quantities of coal to produce thermal power, its major power generation source. When pulverized coal is burned to generate heat, the residual product contained up to 80 per cent fly ash. The fly ash from the exit flue gas was collected at various stages of the flue gas path and at the dust collectors fitted before the final chimney. Market trends revealed that coal would continue to be used as the prime fuel for many more years, resulting in a great amount of fly ash generation.

One challenge for the disposal of the fly ash produced was the need for a massive amount of land, since it was usually dumped on the ground as a waste material. It was estimated that the land requirement for this dumping would be nearby 1,000 square kilometres of land by the year 2015.² Looking at the vast volumes of fly ash produced, and the fact that Indian coal contained high ash content (to the level of 30 to 50 per

¹ Rs1.00 = US\$0.016.

² Department of Forests, Ecology & Environment, Government of Karnataka, “Utility Bonanza from Dust,” *ENVIS Newsletter*, Vol. 2, No. 6, 2007, <http://parisaramahiti.kar.nic.in/11.Fly%20ash%202007—01.pdf>, accessed August 15, 2013.

cent), it was necessary to devise effective uses for it. Accordingly, fly ash was analyzed for its appropriateness in various uses such as manufacturing of ceramic tiles, bricks, building materials, etc.

Fly ash as a construction material was considered appropriate on two accounts. First, there were environmental concerns about the traditional way of making bricks from clay that comprised topsoil. Removing this top layer makes the land infertile for a long period. Using fly ash for making bricks instead of clay would thus help preserve the fertility of the soil. Second, it was estimated that there would be a substantial shortfall in the availability of different types of building materials, including bricks.³ With the country growing at such a rapid pace, the government was keen on promoting fly ash bricks in the construction sector. This would enable a waste product to be used as a construction material and also conserve the environment and resources.

MANUFACTURING PROCESS

The National Thermal Power Corporation (NTPC) of India had developed technology to replace burnt clay bricks with fly ash as a construction material for building walls. The manufacturing process for these bricks, known as fly ash-lime-gypsum bricks, required intimate mixing of fly ash, sand, lime and gypsum. Gypsum and lime were first ground to fine particles and then fly ash and sand added to make a fine blend. The ratio of the input material was as follows:

- Fly ash: 60 to 80 per cent
- Sand: 10 per cent
- Gypsum: 10 per cent
- Lime: 10 to 20 per cent

Water was added to the mix to form a paste after which the mixture was transferred to moulds fitted in a hydraulic/mechanical press. The bricks were later dried in the open for one or two days and then cured using water. A process outline for manufacturing fly ash bricks is given in Exhibit 1.

NTPC established pilot ash brick manufacturing plants at its 13 thermal power plants. Approximately 150 million bricks had already been manufactured in these plants and used for in-house consumption. To guarantee the quality of manufactured bricks, it was suggested that the new brick manufacturing plants follow the guidelines of the quality standard of IS 12894:1990.⁴

SHARMA'S BRICK PROJECT

Sharma was trying to convince Gupta to agree to his business idea. To strengthen his argument, Sharma referred to a portion of the address of Dr. Abdul Kalam, a former Indian president, who had met concerns over the disposal of fly ash by saying:

... the use of coal for power generation results in increased quantum of fly ash production, which has reached over 100 million tonnes per year. All-out efforts are needed to utilize this fly ash not only for environmental considerations, but also to avoid land usage for fly ash dumping. Though there has been a steady progress in fly ash utilization from 1990, we have a long way to go to reach the target of 100 per cent fly ash utilization. It is reported that the agricultural increase of grains is around 15 per cent, green vegetables 35 per cent and root vegetables 50 per cent when

³ The Ministry of Statistics and Programme Implementation, Government of India, "Housing," http://mospi.nic.in/Mospi_New/upload/SYB2013/CH—28—HOUSING/HOUSING%20—%20WRITEUP.pdf, accessed August 15, 2013.

⁴ IS 12894:1990 is the Indian standard for fly ash—lime bricks. www.bis.org.in/index.asp, accessed August 15, 2013.

fly ash is mixed with the soil. Toxicity tests have proved that there is no toxic element due to fly ash. But it has higher nutrients due to increased availability of iron and calcium. The fly ash can become a wealth generator by making use of it for agriculture and producing “green building” materials. Presently, the fly ash utilization is in the range of 40 million tonnes per year providing employment for over 50,000 personnel. At the full utilization of the generating stock, this will provide employment potential for 300,000 people and result in a business volume of over Rs40,000 million per year.⁵

After considering this speech, the two friends continued to discuss the project.

Gupta: I think it's a great idea to use fly ash to manufacture bricks. The availability of fly ash, I guess, will never be a problem. Nearly three-fourths of the installed power generation capacity of India comes from thermal plants, in 90 per cent of which coal is used as the primary fuel. As long as coal is used to generate thermal power, the residual product — fly ash — will continue to be produced.

Sharma: Exactly. You *do* see there is a huge business potential. The current prevalent method of its disposal, primarily dumping, requires large areas of land. On top of that, currently the thermal power plants are generating 100 million tonnes of fly ash per year in India, and such an enormous quantity is creating difficulties for disposal and an ecological menace. According to an estimate that I read, dumping of fly ash would need 1,000 square kilometres of land by 2015. Looking at such potential hazards, the central and state governments are trying to encourage such units. This could be a good opportunity for us. Think of it, fly ash is available in huge quantities at a cheap rate, the government is enthusiastic about it and there is a huge demand for it, plus we have experience and contacts in the construction sector. What more can anyone want? What do you say?

Gupta: Yes . . . you were doing some research on its market potential?

Sharma: Yes, I did that. Ok. Let me show you the market potential.

Sharma then presented a power point display of the following points.

Market Potential

The annual demand for clay bricks for construction purposes was estimated to be 180 billion tonnes that required almost 340 billion tonnes of clay annually. Production of clay bricks required clay that resulted in leeching the fertility of the land in the long run. With construction progressing at a very fast pace, the government was keen to develop a regular supply of bricks without causing damage to the soil.

The use of fly ash in such circumstances was a priority. The quality and durability of fly ash brick were considered very suitable for application in various construction projects. A large quantity of fly ash was available as waste material from thermal power plants. The government was encouraging manufacturing of fly ash bricks, and various concessions were given to such manufacturers. Further, these bricks were factory-made by mixing the ingredients with water without consuming thermal energy.

Manufacturers of construction materials had also begun to note the importance and application of fly ash bricks. They knew that with an increase in construction activities, there would be an equally rising

⁵ “Address during the Inauguration of Workshop on Campaign Clean India by the Ministry of Tourism,” December 20, 2011, www.abdulkalam.com/kalam/jsp/display_searchcontent.jsp, accessed August 15, 2013.

demand for bricks. The housing sector was predicted to experience a shortage of 20 million to 70 million home units⁶, which presented a ripe market for the use of cheap fly ash as a raw material in construction.

On the basis of such preliminary analysis, Sharma decided to set up a plant that would have the capacity to manufacture four million bricks. He presented the calculations for discussion.

Investments Required

Sharma: Alok, at the outset we require Rs8 million as an initial investment in fixed assets. Here, take a look at the costing [he hands over a sheet of paper. According to my estimate, the major expenses would be in transport vehicles and machinery. The building modifications will require approximately Rs1.40 million. The water supply arrangements will be another minor expense.

Gupta examined the table illustrating the estimated expenses (see Exhibit 2).

Sharma: In addition to the above investments, I have estimated the working capital requirements, which are expected to be approximately Rs2 million. Other routine expenses are estimated as follows [see Exhibit 3]. Have a look. [He hands over another sheet of paper].

By the way, there would be the other operating expenses related to the volume of production. These are essentially the raw materials, electricity and labour for making the final product. According to my estimates, the cost of major raw materials, energy requirements and labour would be Rs900,000 per month (see Exhibit 4).

Sharma handed the estimates of the manpower costs to Gupta (see Exhibit 5). Sharma himself would work as production manager, since hiring another person for the post would cost them Rs50,000 per month.

Gupta was puzzled and wanted to ascertain more clearly the feasibility of the project. They agreed to consult expert to evaluate their business plan.

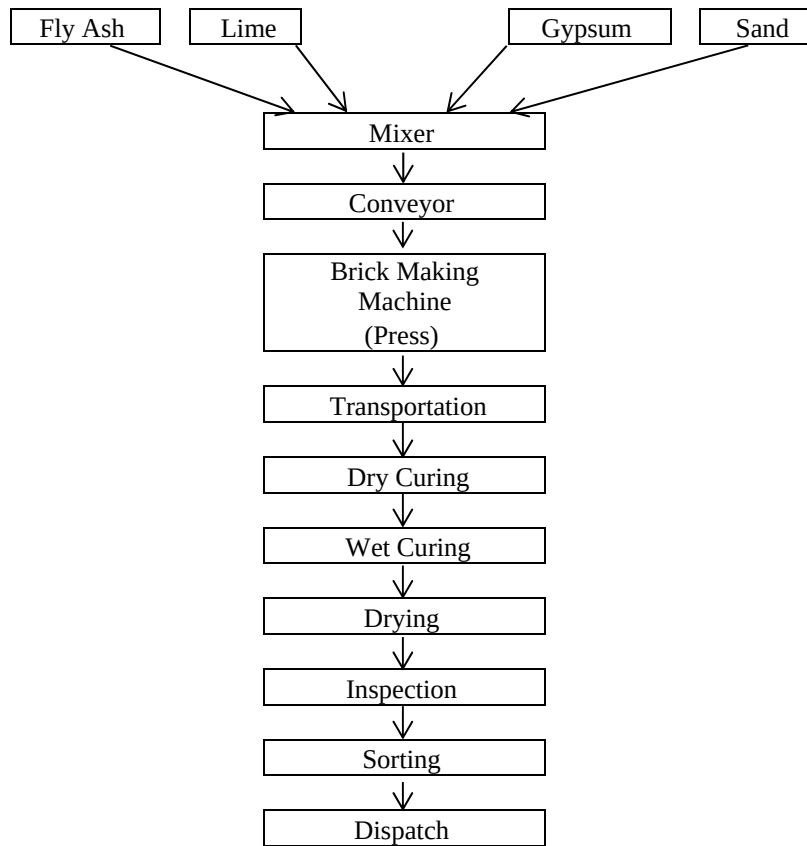
FINAL DECISION

A detailed discussion followed the conversation. Although the proposed plant had the capacity to produce four million bricks per year, actual production would depend on market demand. Further, output would decline in case of any breakdowns of plant and equipment. Finally, the partners estimated that a sales volume of 2.4 million bricks could be sold per year at an average price of Rs7,000 per 1,000 bricks. The initial investment would cost them Rs10 million, out of which Rs6 million would be invested by the partners from their own resources. A local bank had agreed to provide a loan for the balance at an interest rate of 12 per cent per annum against the mortgage of the equipment. Sharma would work full time in the business and would draw a salary of Rs50,000 per month. The life of the project was estimated to be five years. The salvage value of the plant and equipment at the end of five years would be negligible and could be ignored. However, investments in the working capital would be recovered in full.

Before actually investing, the partners requested a consultant to review their business plan.

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⁶ The World Bank, "Expanding Housing Finance to the Underserved in South Asia: Market Review and Forward Agenda," 2010, <http://siteresources.worldbank.org/SOUTHASIAEXT/Resources/223546—1269620455636/6907265—1284569649355/CompleteReportSARHousingFinanceOctober2010.pdf>, accessed August 15, 2013.

EXHIBIT 1: FLY ASH BRICK MANUFACTURING PROCESS

Source: Compiled by authors.

EXHIBIT 2: ESTIMATED INVESTMENT IN INDIAN RUPEES

Building Modification	1,400,000
Water supply arrangements	100,000
Machinery	2,000,000
Trucks	3,000,000
Payload machine	1,500,000
Total	8,000,000

Source: Compiled by authors.

EXHIBIT 3: ROUTINE EXPENSES PER MONTH IN INDIAN RUPEES

Building rent	50,000
Administrative cost	10,000
Office supply	5,000
Electricity (for lighting)	10,000
Miscellaneous	20,000

Source: Compiled by authors.

**EXHIBIT 4: EXPENSES RELATED TO VOLUME OF PRODUCTION PER MONTH IN INDIAN RUPEES
(FOR 0.20 MILLION BRICKS)**

Fly ash	250,000
Gypsum	250,000
Lime	300,000
Sand	40,000
Electricity	10,000
Labour	50,000
Total	900,000

Source: Compiled by authors.

EXHIBIT 5: PERSONNEL COSTS PER MONTH IN INDIAN RUPEES

Workers	100,000
Office assistant	20,000
Watchman	15,000
Drivers	25,000

Source: Compiled by authors.